Eigenvalues and Eigenvectors

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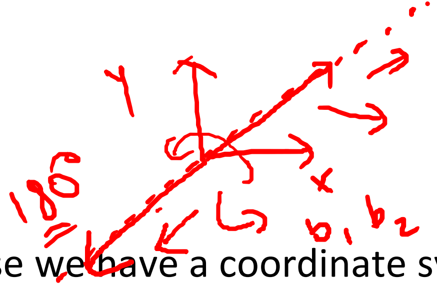
Recap

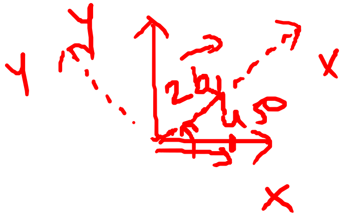
- Matrices
- Vectors
- Basis and unit vectors
- Linear combination of vectors
- Linear Transformations
- Change of Basis → ref. → new coordinate

↳ Linear transformation

Change of Basis

Eigenvectors and Eigenvalues





$\vec{V} = 2\hat{i}$

$\vec{V} \rightarrow$ x-y ref. $\hat{i}, \hat{j}$

Suppose we have a coordinate system and we translate it to a new coordinate space.

- Majority of the vectors $\vec{V}$ from the original space will be knocked off from their span.
- **However, there exists some special set of vectors which remain on their span and are scaled by some factor.**

knocked off

These set of vectors are known as ***Eigenvectors***.

The scaling of the eigenvectors is determined by ***Eigenvalues***. $\rightarrow$ +ve

-ve eigenvalue

-ve

E.X.

$\lambda = 1$

$A = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix}$

Eigenvectors $\begin{bmatrix} -1 \\ 1 \end{bmatrix} \rightarrow \lambda > 1 \rightarrow$

$0 < \lambda < 1 \rightarrow$ compressed

$\lambda < 0$ $\quad$ λ -ve

Eigenvalue $\rightarrow 2$

$\boxed{\vec{V} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}}$

factor of 2

$\vec{V} = \begin{bmatrix} 3 \\ -1 \end{bmatrix} \rightarrow \alpha \cancel{\begin{bmatrix} 3 \\ -1 \end{bmatrix}}$

Mathematical Definition

Mathematically eigenvectors and eigenvalues can be obtained from:

$$A\vec{V} = \lambda\vec{V}$$

Where,

A is the transformation matrix

$\vec{V}$ is the eigenvector

λ is the eigenvalue

$A \rightarrow \vec{V}$ ↳ eg.

Some set of vectors

2x2

matrix x vector = Scaler x vector

$\lambda [\vec{V}]$

$\lambda \vec{V} = \lambda I \vec{V}$

$A\vec{V} = \lambda I \vec{V}$

$A\vec{V} - \lambda I \vec{V} = 0$

$\vec{V} \neq 0$

or $(A - \lambda I)\vec{V} = 0$

Change of basis

Linear tran.

→ origin → remains on it's posit.

$(A - \lambda I)\vec{V} = 0$

↳ non-zero

$\rightarrow A - \lambda I = 0$

↳ $\lambda \rightarrow$ Eigenvalues

Eigenvectors and Eigenvalues

E.X. $A = \begin{bmatrix} 2 & 2 \\ 1 & 3 \end{bmatrix}$

$A\vec{v} = \lambda I \vec{v}$

$\det(A - \lambda I) = 0$

$\lambda \to$ eigenvalues

$$\begin{bmatrix} 2 & 2 \\ 1 & 3 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = 0$$

$$\Rightarrow \begin{bmatrix} 2 & 2 \\ 1 & 3 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} = 0$$

$$\det \begin{bmatrix} 2-\lambda & 2 \\ 1 & 3-\lambda \end{bmatrix} = 0$$

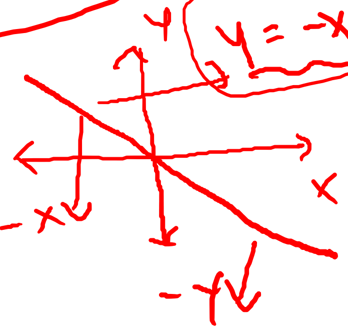
$(2-\lambda)(3-\lambda) - (2)(1) = 0$

$\lambda^2 - 5\lambda + 4 = 0$

$\lambda = 1$ & $\lambda = 4$

$\det(A - \lambda I) = 0$

Set 1, 4



Ex. $A = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix}$

$\lambda = 2$ or $\lambda = 3$

Eigen vectors

$(A - \lambda I)\vec{V} = 0$

$$\begin{bmatrix} 3-2 & 1 \\ 0 & 2-2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

$$\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

$$\begin{bmatrix} x+y \\ 0 \end{bmatrix} = 0$$

$x + y = 0$

$y = -x$

Eigenvectors and Eigenvalues

E.X.

$$A = \begin{bmatrix} 2 & 0 & 0 \\ -1 & 1 & 0 \\ 5 & 3 & -3 \end{bmatrix}$$

3×3

$\det(A - \lambda I) = 0$

$$\begin{bmatrix} 2 & 0 & 0 \\ -1 & 1 & 0 \\ 5 & 3 & -3 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\det \begin{bmatrix} 2-\lambda & 0 & 0 \\ -1 & 1-\lambda & 0 \\ 5 & 3 & -3-\lambda \end{bmatrix} = 0$$

$(2-\lambda)\left[(1-\lambda)(-3-\lambda) - (3)(0)\right]$

$(2-\lambda)(1-\lambda)(-3-\lambda) = 0$

$-3-\lambda = 0$

$-3 = \lambda$

$$\boxed{\begin{aligned} \lambda &= 1 \\ \lambda &= 2 \\ \lambda &= -3 \end{aligned}}$$

3 eigenvalues

3 sets of eigenvectors

$$(A - \lambda I)\begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$

~~Eigenvectors~~

$$A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

$\lambda = \sqrt{-1}$

$i \,/\, j$

~~unit~~

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

$\lambda = 0, \ \lambda = 2$

$A = [b_1 \; b_2]$

$\hat{i} \; \hat{j}$

E.X.

$\hat{i} = \begin{bmatrix} -1 \\ 0 \end{bmatrix} \quad \hat{j} \quad \begin{bmatrix} 0 \\ 2 \end{bmatrix}$

$\vec{b}_1 \quad \vec{b}_2$

$y_2 \qquad \vec{x}^{\,-1}$

eigen vectors

Eigenbasis

$A = \begin{bmatrix} -1 & 0 \\ 0 & 2 \end{bmatrix}$

$\lambda = -1, \; \lambda = 2$

If the basis vectors are eigenvectors

$A \to$ diagonal $\quad \lambda$

$A = \begin{bmatrix} -5 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -13 \end{bmatrix}$

$4 \to$

Series of transform $\to \boxed{} \to$ editing

$\underbrace{\begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix} \cdot \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix} \cdot \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix} \cdots}_{n} \begin{bmatrix} x \\ y \end{bmatrix} =$

$\begin{bmatrix} 3^n & 0 \\ 0 & 2^n \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 3^n x \\ 2^n y \end{bmatrix}$

Thank you!